Roughly 10^{-35} seconds after the burst began, the inflaton field found its way off the high-energy plateau and its value throughout space slid down to the bottom of the bowl, turning off the repulsive push. And as the inflaton value rolled down, it relinquished its pent-up energy to the production of ordinary particles of matter and radiation—like a foggy mist settling on the grass as morning dew—that uniformly filled the expanding space.[14] From this point on, the story is essentially that of the standard big bang theory: space continued to expand and cool in the aftermath of the burst, allowing particles of matter to clump into structures like galaxies, stars, and planets, which slowly arranged themselves into the universe we currently see, as illustrated in Figure 10.3.

Guth's discovery—dubbed *inflationary cosmology*—together with the important improvements contributed by Linde, and by Albrecht and Steinhardt, provided an explanation for what set space expanding in the first place. A Higgs field perched above its zero energy value can provide an outward blast driving space to swell. Guth provided the big bang with a bang.

The Inflationary Framework

Guth's discovery was quickly hailed as a major advance and has become a dominant fixture of cosmological research. But notice two things. First, in the standard big bang model, the bang supposedly happened at time zero, at the very beginning of the universe, so it is viewed as the creation event. But just as a stick of dynamite explodes only when it's properly lit, in inflationary cosmology the bang happened only when conditions were right— when there was an inflaton field whose value provided the energy and negative pressure that fueled the outward burst of repulsive gravity—and that need not have coincided with the "creation" of the universe. For this reason, the inflationary bang is best thought of as *an* event that the preexisting universe experienced, but not necessarily as *the* event that created the universe. We denote this in Figure 10.3 by maintaining some of the fuzzy patch of Figure 9.2, indicating our continuing ignorance of fundamental origin: specifically, if inflationary cosmology is right, our ignorance of why there is an inflaton field, why